

Development of Punctuation Sensitivity in Early Readers

An ERP Investigation of the Closure Positive Shift in Grades 3–5

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1 · Background

- Readers may treat commas as cues to implicit prosody that help them build syntactic structure; punctuation knowledge, linked to prosodic awareness, predicts reading comprehension [1].
- In adults, the closure positive shift (CPS) marks prosodic boundaries in speech [2] and commas in silent reading, and is larger in readers who follow comma rules strictly [3].
- The speech CPS is present by age 6 [4]; whether commas elicit a CPS in developing readers is unknown.
- Grades 3–5 span the transition to fluent reading, as punctuation awareness grows [1] and syntactic skills increasingly support comprehension [5].

2 · Questions

- Q1 Do correct commas elicit a CPS?**
Correct vs Missing comma (same word, ± comma)
- Q2 Are misplaced commas detected?**
Incorrect vs Correct comma
- Q3 Do effects track reading skill or age?**
WRMT-PC, age and punctuation tasks

3 · Participants

Grade	n (F)	Age	WRMT-PC	PK	P-SR
3	8 (4)	8.7	16.9	14.0	0.22
4	4 (4)	10.1	24.8	18.8	0.71 ^a
5	6 (3)	10.8	22.8	16.3	0.57
All	18 (11)	9.7	20.6	15.8	0.41 ^b

Means. Age in years. WRMT-PC: WRMT-III Passage Comprehension (raw). PK: Punctuation Knowledge; P-SR: proportion of punctuation marks noticed while silently reading a story (lab tasks). ^an = 2; ^bn = 16. Also given: TOWRE-2, CTOPP-2 Elision, WRMT-III Word Comprehension, WASI-II Matrix Reasoning.

References

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4 · Stories & design

Children silently read three stories (101 sentences) word by word, with a picture question after each section. Across two counterbalanced lists, each sentence appeared intact or with its comma moved:

ORIGINAL

Since Tim loved adventure, he sat up front in the bus.

— Correct comma · 44.5 trials (34–48)

COMMA MOVED

Since Tim loved adventure he, sat up front in the bus.

--- Missing comma · 29 (19–30)
— Incorrect comma · 33 (28–37)

Median trials per child after artifact rejection (range).

Tim

loved

adventure,

he

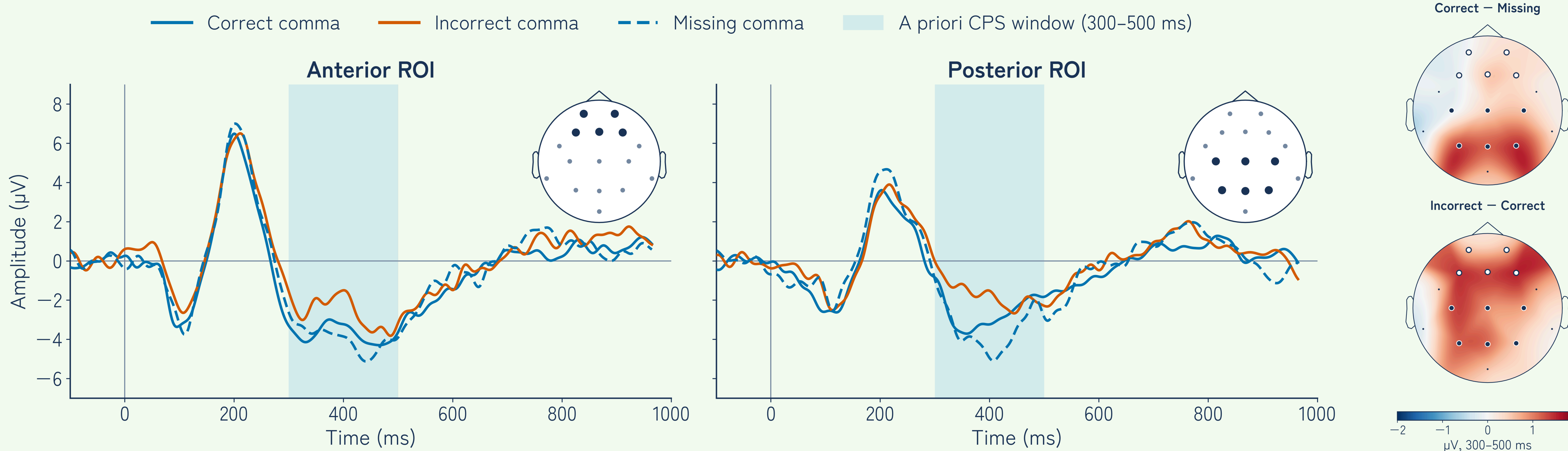
sat

One word every 1000 ms at screen centre; ERPs time-locked to critical-word onset (−100 to 1000 ms).

5 · EEG & analysis

- 16 active electrodes + 2 EOG (actiCAP, V-Amp; 500 Hz), filtered 0.1–30 Hz.
- ICA (Picard) removed ocular components; AutoReject repaired or rejected noisy epochs; trials in which any ROI electrode exceeded ±100 µV at any point in the epoch were rejected (1.1%).
- Epochs −100 to 1000 ms from critical-word onset (corrected for a measured 36 ms display lag); mastoid reference; baseline −100 to 0 ms.
- Mean amplitude 300–500 ms (a priori CPS window) over anterior (Fp1, Fp2, F3, Fz, F4) and posterior (C3, Cz, C4, P3, Pz, P4) ROIs.
- Linear mixed-effects model (lme4) on ROI-averaged single trials: Condition × ROI, with random intercepts for child, child × condition, child × ROI, child × condition × ROI and trial. Contrasts within each ROI, Holm-corrected across the four tests; child-level *t*-tests as a check.

6 · Results



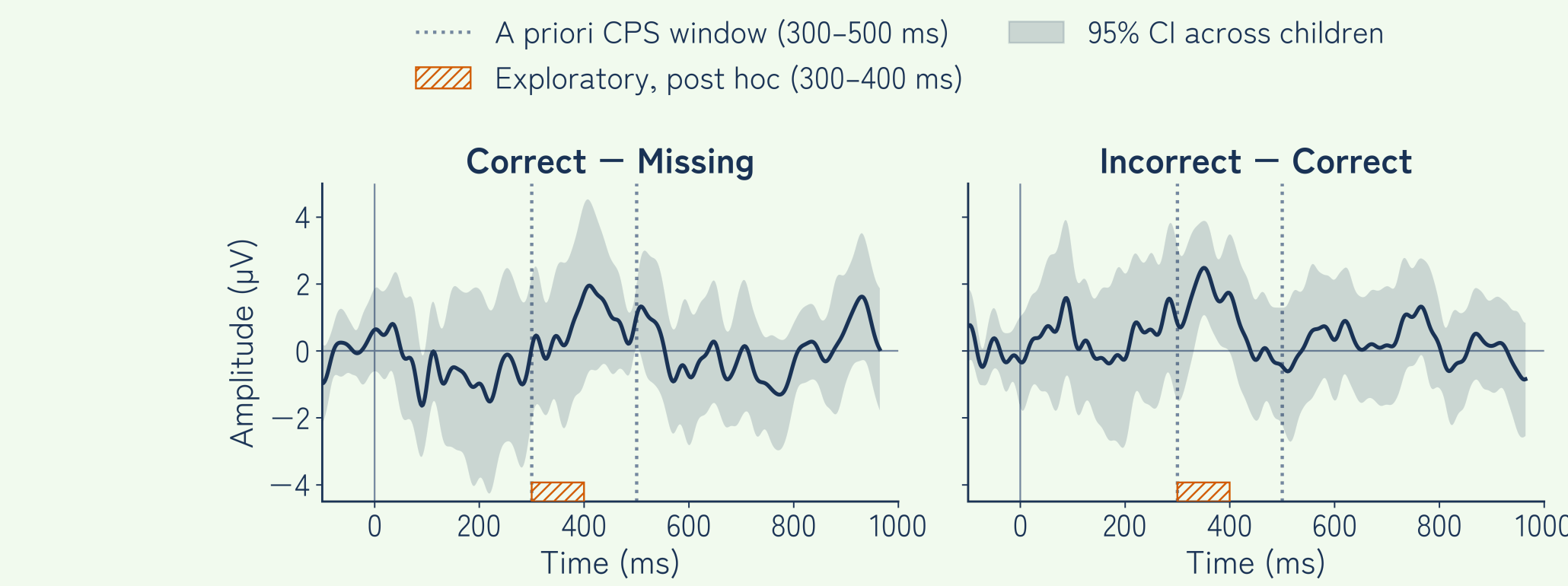
Planned contrasts, 300–500 ms

ROI	Contrast	Δ µV [95% CI]	p	p Holm	Children > 0
Anterior	Correct — Missing	0.16 [−1.27, 1.60]	.82	.82	10/18
Anterior	Incorrect — Correct	1.04 [−0.33, 2.41]	.14	.55	12/18
Posterior	Correct — Missing	0.72 [−0.72, 2.15]	.32	.65	10/18
Posterior	Incorrect — Correct	0.91 [−0.46, 2.29]	.19	.57	12/18

Mixed-model estimates (Satterthwaite df). Children > 0: children whose own mean difference was positive. Child-level *t*-tests agreed (all *p* ≥ .08).

Update to the abstract: the abstract's results (n = 18) used an aggressive outlier trim (±2 SD of single-channel values) and channel-level models. With artifact-based rejection and trial-level models, neither effect is reliable.

7 · Exploratory

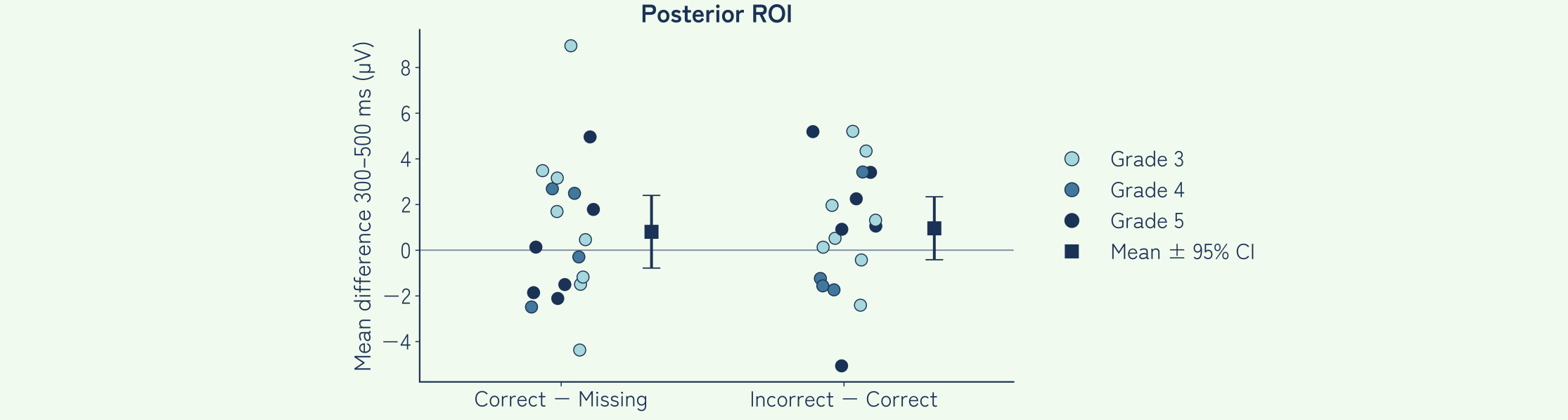


Posterior ROI. Incorrect — Correct peaked early in the window, so we examined 300–400 ms post hoc (hatched).

Posterior Incorrect — Correct: +1.62 µV [0.13, 3.10], *p* = .033 (Holm *p* = .13); 13 of 18 children > 0. Anterior: +1.27 µV [−0.22, 2.75], *p* = .095. Correct — Missing: n.s. at both ROIs (*p* ≥ .69).

Caveat: window chosen from these data, so *p*-values are biased low; Incorrect and Correct commas also fall on different words. A hypothesis to test in new data.

8 · Individual differences



One dot per child (posterior ROI, 300–500 ms); squares: mean ± 95% CI.

Brain-behaviour (child-level regressions, N = 18). Reading comprehension (WRMT-PC, controlling age) did not predict either effect at either ROI (lpartial *r* ≤ .29, all *p* ≥ .26), nor did age (all *p* ≥ .24) or the punctuation tasks (exploratory; all *p* ≥ .07).

Where the variance lies (SDs, µV). Single trials are very noisy: SD 12.2 µV from trial to trial (pooled across children). Children differ more in the amplitude of each condition (SD 1.7 µV between children) than in the differences between conditions (SD 0.8 µV): a child with large ERPs has them in every condition, so subtraction removes most of what distinguishes children. With ~30–45 trials per condition, each child's effect has a standard error of ~2.9 µV, over three times that 0.8 µV spread, so individual effects cannot yet be measured reliably, let alone related to reading skill (cf. generalizability theory [6]).

Interim findings

- Q1** No reliable comma CPS in grades 3–5 so far: correct and missing commas did not differ at 300–500 ms.
- Q2** Misplaced commas may evoke an early positivity (300–400 ms), a post hoc effect to test in new data.
- Q3** Variability lies mostly within children, not between them, so individual differences are not yet measurable: the next phase needs more critical trials per child.